

## 45.4 Expectile regression

The check loss buys freedom from distributional assumptions at the price of a kink, and the kink makes the estimator awkward: no closed form, no hat matrix, no leverage shortcut for cross-validation (Theorem 29.3.2), no easy way to attach the quadratic penalties that [Chapter 30](#) and [Chapter 43](#) are built on. Newey and Powell (1987) replaced the absolute deviation by a squared one, keeping the asymmetry. The answer is a family of location measures called expectiles, fitted by weighted least squares.

### 45.4.1 Asymmetric squared loss

#### Definition 45.4.1 — Expectiles and expectile regression

For  $\tau \in (0, 1)$  define the **asymmetric squared loss**

$$\rho_\tau^E(u) = |\tau - 1\{u < 0\}| u^2 = \begin{cases} \tau u^2, & u \geq 0, \\ (1 - \tau)u^2, & u < 0. \end{cases} \quad (45.4.1)$$

For a random variable  $Y$  with  $EY^2 < \infty$ , the  $\tau$ -**expectile**  $\mu_\tau$  is a minimizer of  $S(\mu) = E\rho_\tau^E(Y - \mu)$ . The **expectile regression** model at level  $\tau$  asserts that the  $\tau$ -expectile of the conditional law of  $Y$  is  $\mathbf{x}^\top \boldsymbol{\beta}_\tau$ , and the estimate is

$$\hat{\boldsymbol{\beta}}_\tau^E \in \arg \min_{\mathbf{b}} \sum_{i=1}^n \rho_\tau^E(y_i - \mathbf{x}_{(i)}^\top \mathbf{b}). \quad (45.4.2)$$

The weight  $\tau$  applies above the fit and  $1 - \tau$  below, as in the check loss; only the power has changed, and at  $\tau = 1/2$  problem [Eq. 45.4.2](#) is least squares.

### 45.4.2 What an expectile is

#### Proposition 45.4.1 — Properties of expectiles

Let  $EY^2 < \infty$  and let  $Y$  be non-degenerate.

- (a) **Existence, uniqueness, monotonicity.**  $S$  is finite, differentiable and strictly convex, and  $\mu_\tau$  is its unique stationary point, characterized by

$$\tau E(Y - \mu_\tau)_+ = (1 - \tau) E(\mu_\tau - Y)_+. \quad (45.4.3)$$

The map  $\tau \mapsto \mu_\tau$  is strictly increasing on  $(0, 1)$ .

- (b) **The mean is an expectile.**  $\mu_{1/2} = EY$ .
- (c) **The level of a point.** For every  $\mu$  with  $P(Y < \mu) > 0$  and  $P(Y > \mu) > 0$  — equivalently, every  $\mu$  interior to the convex hull of the support of  $Y$  — there is exactly one  $\tau$  with  $\mu_\tau = \mu$ , namely

$$\tau = \frac{E(\mu - Y)_+}{E|Y - \mu|}. \quad (45.4.4)$$

- (d) **Equivariance.** For  $a > 0$  and  $b \in \mathbb{R}$ ,  $\mu_\tau(aY + b) = a\mu_\tau(Y) + b$ , and  $\mu_\tau(-Y) = -\mu_{1-\tau}(Y)$ . Expectiles are *not* equivariant under nonlinear increasing transformations.

*Proof.*

- (a) Write  $S(\mu) = \tau E(Y - \mu)_+^2 + (1 - \tau) E(\mu - Y)_+^2$ , finite because  $(y - \mu)^2 \leq 2y^2 + 2\mu^2$ . The loss is continuously differentiable with  $\rho_\tau^{E'}(u) = 2u|\tau - 1\{u < 0\}|$ , continuous at the origin because both one-sided derivatives vanish, and nondecreasing, so  $\rho_\tau^E$  is convex. Dominated convergence permits differentiation under the expectation, giving

$$\frac{1}{2}S'(\mu) = -\tau E(Y - \mu)_+ + (1 - \tau) E(\mu - Y)_+ =: -H(\mu).$$

Now  $H$  is continuous, and for  $\mu < \mu'$ ,

$$\begin{aligned} H(\mu) - H(\mu') &= \tau E\{(Y - \mu)_+ - (Y - \mu')_+\} \\ &\quad + (1 - \tau) E\{(\mu' - Y)_+ - (\mu - Y)_+\} \geq 0, \end{aligned}$$

each expectation being nonnegative because  $t \mapsto (y - t)_+$  is nonincreasing and  $t \mapsto (t - y)_+$  nondecreasing, and strict unless  $Y$  puts no mass outside  $[\mu, \mu']$  on either side, which a non-degenerate  $Y$  cannot do for every pair. Moreover, as  $\mu \rightarrow -\infty$  monotone convergence gives  $E(\mu - Y)_+ \downarrow 0$ , while  $E(Y - \mu)_+ \geq EY - \mu \rightarrow +\infty$ , so  $H(\mu) \rightarrow +\infty$ ; the same argument reflected gives  $H(\mu) \rightarrow -\infty$  as  $\mu \rightarrow +\infty$ . So  $H$  has a unique zero,  $S$  is strictly convex there, and Eq. 45.4.3 is exactly  $H(\mu_\tau) = 0$ . For monotonicity, let  $\tau < \tau'$  and write  $H_\tau$  for the function above. Then

$$\begin{aligned} H_{\tau'}(\mu_\tau) &= H_\tau(\mu_\tau) + (\tau' - \tau) E|Y - \mu_\tau| \\ &= (\tau' - \tau) E|Y - \mu_\tau| > 0, \end{aligned}$$

using  $E(Y - \mu)_+ + E(\mu - Y)_+ = E|Y - \mu|$ ; since  $H_{\tau'}$  is decreasing with a unique zero, that zero  $\mu_{\tau'}$  lies to the right of  $\mu_\tau$ .

- (b) At  $\tau = 1/2$ , Eq. 45.4.3 reads  $E(Y - \mu)_+ = E(\mu - Y)_+$ , that is  $E(Y - \mu) = 0$ .
- (c) Put  $N = E(Y - \mu)_+$  and  $P = E(\mu - Y)_+$ ; both are strictly positive under the stated condition on  $\mu$ , so  $N + P = E|Y - \mu| > 0$ . Equation Eq. 45.4.3 says  $\tau N = (1 - \tau)P$ , hence  $\tau(N + P) = P$ , and Eq. 45.4.4 follows. Conversely, given such a  $\mu$ , the ratio  $P/(N + P)$  lies strictly between 0 and 1 and solves the equation, and by (a) the solution for that  $\tau$  is unique.
- (d) For  $a > 0$ ,  $\rho_\tau^E(au) = a^2 \rho_\tau^E(u)$ , so  $E \rho_\tau^E(aY + b - \mu) = a^2 E \rho_\tau^E\{Y - (\mu - b)/a\}$  and the minimizers correspond as claimed;  $\rho_\tau^E(-u) = \rho_{1-\tau}^E(u)$  gives the reflection. For the failure under nonlinear maps it is enough to look at  $\tau = 1/2$ : by (b) the 1/2-expectile is the mean, and for strictly convex  $h$  Jensen's inequality gives  $Eh(Y) > h(EY)$ , so  $\mu_{1/2}\{h(Y)\} \neq h\{\mu_{1/2}(Y)\}$ .

■

Part (c) says what level an expectile sits at. Compare the two characterizations:

$$\tau = \frac{E \mathbf{1}\{Y \leq q_\tau\}}{E \mathbf{1}} \quad (\text{quantile}), \quad \tau = \frac{E(\mu_\tau - Y)_+}{E|Y - \mu_\tau|} \quad (\text{expectile}).$$

A quantile counts the mass below the point; an expectile weighs it, by distance. So an expectile depends on the *entire* distribution and has none of the robustness of Definition 45.2.1: one response moved to  $+10^6$  moves every expectile. In exchange it is smooth.

*Remark* (Expectiles as quantiles of another law). Since Eq. 45.4.4 determines  $\tau$  from  $\mu$  and  $F$ , the expectiles of  $F$  are the quantiles of another distribution built from  $F$  by partial moments — the observation of Jones (1994). The catch is that the relabelling depends on  $F$ : the  $\tau$ -quantile is at level  $\tau$  for every distribution, while the  $\tau$ -expectile is at a level only the distribution can tell you.

### Example. (Which quantile is the nine-tenths expectile?)

For a standard normal the 0.9-expectile sits at the 0.806-quantile and the 0.5-expectile at the 0.500-quantile, as symmetry requires. For a lognormal with log-scale 0.8 the 0.9-expectile sits at the 0.892-quantile: the heavy right tail drags it further out. For the residuals of the least squares fit to Engel's budgets the 0.9-expectile sits at the 0.851-quantile and the 0.1-expectile at the 0.170-quantile (Figure 45.4.1(b)). Three distributions, three answers to the same question.

## 45.4.3 Fitting by weighted least squares

### Proposition 45.4.2 — The estimating equation and its iteration

Let  $\text{rank}(\mathbf{X}) = p$  and  $Q(\mathbf{b}) = \sum_i \rho_\tau^E(y_i - \mathbf{x}_{(i)}^\top \mathbf{b})$ .

- (a)  $Q$  is continuously differentiable and strictly convex, so it has a unique minimizer, characterized by

$$\mathbf{X}^\top \mathbf{W}(\mathbf{b}) (\mathbf{y} - \mathbf{X}\mathbf{b}) = \mathbf{0}, \quad \mathbf{W}(\mathbf{b}) = \text{diag}\{w_i(\mathbf{b})\}, \quad w_i(\mathbf{b}) = |\tau - 1\{y_i < \mathbf{x}_{(i)}^\top \mathbf{b}\}|. \quad (45.4.5)$$

- (b) The **iterated weighted least squares** step  $\mathbf{b}^{(t+1)} = \{\mathbf{X}^\top \mathbf{W}(\mathbf{b}^{(t)}) \mathbf{X}\}^{-1} \mathbf{X}^\top \mathbf{W}(\mathbf{b}^{(t)}) \mathbf{y}$  has the property that if  $\mathbf{W}(\mathbf{b}^{(t+1)}) = \mathbf{W}(\mathbf{b}^{(t)})$ , then  $\mathbf{b}^{(t+1)}$  is the exact global minimizer of  $Q$ .

*Proof.*

- (a) The scalar loss is continuously differentiable with derivative  $2u|\tau - 1\{u < 0\}|$ , which is strictly increasing, so  $\rho_\tau^E$  is strictly convex; composing with the affine map  $\mathbf{b} \mapsto y_i - \mathbf{x}_{(i)}^\top \mathbf{b}$  and summing gives a convex  $Q$ , strictly convex because  $\text{rank}(\mathbf{X}) = p$  makes  $\mathbf{b} \mapsto \mathbf{X}\mathbf{b}$  injective. It is also coercive:  $\rho_\tau^E(u) \geq \min(\tau, 1 - \tau)u^2$  gives  $Q(\mathbf{b}) \geq \min(\tau, 1 - \tau)\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2$ , which tends to infinity as  $\|\mathbf{b}\| \rightarrow \infty$  because  $\mathbf{X}$  has full column rank, so a minimizer exists, and strict convexity makes it unique. The chain rule gives  $\nabla Q(\mathbf{b}) = -2\mathbf{X}^\top \mathbf{W}(\mathbf{b})(\mathbf{y} - \mathbf{X}\mathbf{b})$ , and for a differentiable convex function a stationary point is a global minimum.
- (b) The weighted least squares step solves  $\mathbf{X}^\top \mathbf{W}_t \mathbf{X} \mathbf{b} = \mathbf{X}^\top \mathbf{W}_t \mathbf{y}$  with  $\mathbf{W}_t = \mathbf{W}(\mathbf{b}^{(t)})$ , so  $\mathbf{X}^\top \mathbf{W}_t (\mathbf{y} - \mathbf{X}\mathbf{b}^{(t+1)}) = \mathbf{0}$ . If  $\mathbf{W}(\mathbf{b}^{(t+1)}) = \mathbf{W}_t$ , this is Eq. 45.4.5 at  $\mathbf{b}^{(t+1)}$ , so  $\nabla Q(\mathbf{b}^{(t+1)}) = \mathbf{0}$  and by (a) the point is the global minimizer.

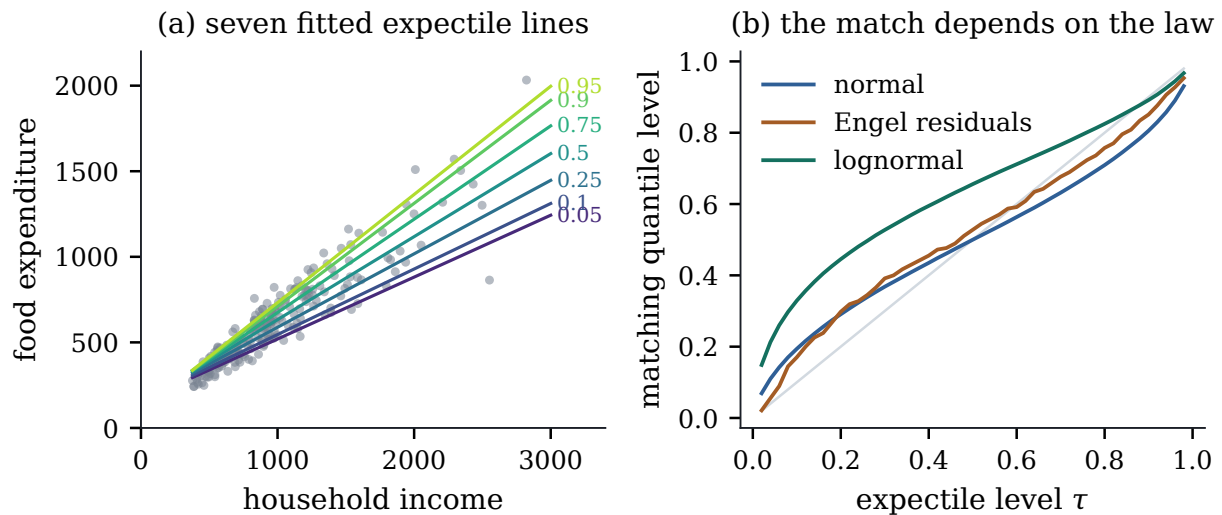


Figure 45.4.1: (a) Seven fitted expectile lines for Engel's budgets. (b) The quantile level at which the  $\tau$ -expectile sits, for three laws; the diagonal is where the levels agree.

*Warning* (What is not proved here). Part (b) says the iteration is *correct when it stops*, not that it stops. The weights depend on  $\mathbf{b}$  only through the signs of the residuals, so the step is a Newton step for a piecewise linear system with finitely many weight patterns; global convergence is not proved in this book. In practice it is fast: for the seven fits below none needed more than 6 reweightings from a least squares start. Compare Proposition 20.6.1, whose majorization argument does give a descent guarantee but requires  $\rho(u) = g(u^2)$  with  $g$  concave, which Eq. 45.4.1 is not.

#### Example. (Expectile lines for Engel's budgets)

The same 235 households, fitted at the same seven levels by Eq. 45.4.2:

| $\tau$    | 0.05   | 0.10   | 0.25   | 0.50   | 0.75   | 0.90   | 0.95   |
|-----------|--------|--------|--------|--------|--------|--------|--------|
| slope     | 0.3617 | 0.3830 | 0.4293 | 0.4852 | 0.5454 | 0.6017 | 0.6322 |
| intercept | 157.9  | 162.6  | 159.0  | 147.5  | 128.8  | 109.0  | 99.8   |

The level  $\tau = 1/2$  reproduces least squares exactly, slope 0.4852. The fan opens in the same direction as the quantile fan of Example (Seven quantile lines for Engel's budgets) but less widely, from 0.3617 to 0.6322 against 0.3434 to 0.7091 — not a discrepancy, since by Example (Which quantile is the nine-tenths expectile?) the level 0.9 means a more central position for an expectile.

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.engel.load_pandas().data
TAUS = (0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.95)

income = data["income"].to_numpy()
food = data["foodexp"].to_numpy()
```

```

n = len(food)
X = np.column_stack([np.ones(n), income])

def expectile_reg(X, y, tau, tol=1e-12, max_iter=100):
    """Expectile regression by iterated weighted least squares.

    The weight of observation  $i$  is  $\tau$  if it sits above the current fit and
     $1 - \tau$  if it sits below; each step is a weighted least squares fit.
    """
    beta, *_ = np.linalg.lstsq(X, y, rcond=None)
    for step in range(1, max_iter + 1):
        w = np.where(y >= X @ beta, tau, 1 - tau)
        new, *_ = np.linalg.lstsq(X * np.sqrt(w)[:, None], y * np.sqrt(w), rcond=
None)
        if np.max(np.abs(new - beta)) <= tol * (1 + np.max(np.abs(beta))):
            return new, step
        beta = new
    raise RuntimeError("iterated weighted least squares did not converge")

fits = {tau: expectile_reg(X, food, tau) for tau in TAUS}
for tau in (0.10, 0.50, 0.90):
    beta, steps = fits[tau]
    print(f"tau = {tau:.2f}: intercept {beta[0]:7.2f} slope {beta[1]:.4f} ({steps}
steps)")

```

[Run this code online](#)

```

def expectile(y, tau):
    """The sample tau-expectile of a vector: the scalar case of the same iteration.
    """
    m = y.mean()
    for _ in range(200):
        w = np.where(y >= m, tau, 1 - tau)
        new = (w * y).sum() / w.sum()
        if abs(new - m) < 1e-12 * (1 + abs(m)):
            return new
        m = new
    return m

def level(y, mu):
    """tau = E(mu - Y)_+ / E|Y - mu|: the expectile level of the point mu."""
    return np.mean(np.maximum(mu - y, 0)) / np.mean(np.abs(y - mu))

ols, _ = np.linalg.lstsq(X, food, rcond=None)[:2]
resid = food - X @ ols
for tau in (0.1, 0.5, 0.9):
    assert abs(level(resid, expectile(resid, tau)) - tau) < 1e-9
print("the identity tau = E(mu - Y)_+ / E|Y - mu| holds at every level tried")

```

[Run this code online](#)

## 45.4.4 When to prefer an expectile

The case for expectiles is about what a differentiable loss makes possible.

**Penalties and smoothing.** A penalized spline turns Eq. 45.4.2 into penalized weighted least squares, which is Definition 30.1.1 with weights. Every step is a linear smoother, so the effective degrees of freedom is a trace and generalized cross-validation Definition 29.3.2 can choose the smoothing parameter through the leverage shortcut Theorem 29.3.2 without refitting. Schnabel and Eilers (2009) built expectile smoothing on this, and Sobotka and Kneib (2012) extended it to structured additive predictors, where the smoothing parameters go to the mixed-model representation (Theorem 43.6.1, using Definition 32.4.1). None of this is available for the check loss without approximation.

**Efficiency and risk.** The expectile estimator uses the magnitude of every residual, the quantile estimator only its sign, whose asymptotic variance carries the sparsity factor of Eq. 45.3.3. Which is the more precise depends on the error law, and the comparison is meaningful only at matched levels, since at a common  $\tau$  the two estimate different functionals; Newey and Powell (1987, §3) give the asymptotic variance from which such comparisons are made. A functional is *elicitable* if it minimizes an expected loss — the mean by squared error, quantiles by the check loss (Theorem 45.2.1), expectiles by Eq. 45.4.1 — and Ziegel (2016) showed that the  $\tau$ -expectiles with  $\tau \geq 1/2$  are the only law-invariant coherent risk measures that are also elicitable, which is what put them into financial risk management.

The case against is short and serious: nobody can say in one sentence what the 0.9-expectile is. “Ninety per cent of households fall below this line” a reader understands; “the lower partial moment at this line is nine times the upper one” is not.

## 45.4.5 Exercises

### 45.4.5.1 A. Check your understanding

#### Exercise A1.

Let  $Y$  take the values 0 and 1 with probability  $1/2$  each. Compute  $\mu_\tau$  from Eq. 45.4.3 as a function of  $\tau$ , and compare with the  $\tau$ -quantile.

*Solution.* For  $\mu \in (0, 1)$ ,  $E(Y - \mu)_+ = (1 - \mu)/2$  and  $E(\mu - Y)_+ = \mu/2$ , so Eq. 45.4.3 gives  $\tau(1 - \mu) = (1 - \tau)\mu$ , that is  $\mu_\tau = \tau$ . The expectile moves continuously from 0 to 1, while the  $\tau$ -quantile jumps from 0 to 1 at  $\tau = 1/2$ . Expectiles smooth over the gaps of a discrete law.

### 45.4.5.2 B. Practice

#### Exercise B1.

For  $Y \sim N(0, 1)$ , show that Eq. 45.4.3 becomes  $\tau\{\varphi(\mu) - \mu(1 - \Phi(\mu))\} = (1 - \tau)\{\varphi(\mu) + \mu\Phi(\mu)\}$ , and verify numerically that  $\mu_{0.9} = 0.8616$  lies at the 0.806-quantile, as reported in Example (Which quantile is the nine-tenths expectile?).

*Solution.* For the standard normal,  $E(Y - \mu)_+ = \varphi(\mu) - \mu\{1 - \Phi(\mu)\}$  and  $E(\mu - Y)_+ = \varphi(\mu) + \mu\Phi(\mu)$ , both by integration by parts. Substituting into Eq. 45.4.3 gives the stated identity; solving it numerically at  $\tau = 0.9$  gives  $\mu_{0.9} = 0.8616$ , and  $\Phi(0.8616) = 0.806$ .

**Exercise B2.**

Show that if the law of  $Y$  is symmetric about  $m$ , then  $\mu_\tau + \mu_{1-\tau} = 2m$ , and that the  $\tau$ -expectile lies at the  $\alpha(\tau)$ -quantile with  $\alpha(1/2) = 1/2$ . Does symmetry force  $\alpha(\tau) = \tau$ ?

**45.4.5.3 C. Going deeper****Exercise C1.**

Let  $Y$  have a Cauchy distribution. Which of the objects in this chapter still exist: the conditional mean, the  $\tau$ -quantiles, the  $\tau$ -expectiles? What does this say about the price of the smooth loss?

*Solution.* Quantiles exist and are finite for every  $\tau \in (0, 1)$ , since Theorem 45.2.1(a) needs only a distribution function (and the remark after it removes even the moment condition). The mean does not exist, so neither does the 1/2-expectile; indeed [Definition 45.4.1](#) assumes  $EY^2 < \infty$ , and  $E|Y| = \infty$  already destroys [Eq. 45.4.3](#), whose two sides are both infinite. The smooth loss buys computation at the cost of moments.